

SAIRAJ PANKAR

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SUMMARY

AI/ML Software Engineer with 1+ years of hands-on experience building and deploying machine learning, computer vision, and real-time AI systems for industrial applications. Strong in Python, PyTorch, YOLO, Scikit-learn, FastAPI, CUDA, and Linux, with experience spanning model development, GPU-accelerated inference, distributed software systems, and automated deployment. Optimized vision pipelines handling 70+ concurrent IP and thermal camera streams, reducing CPU utilization from 80%+ to 35–40% through GPU acceleration and resource optimization.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java
- **AI/ML & Applied AI:** PyTorch, Scikit-learn, YOLO, OpenCV, LightGBM, XGBoost, CatBoost, Pandas, NumPy, Embeddings, Semantic Search, Pose Estimation
- **LLM & Generative AI:** RAG (Retrieval-Augmented Generation), LangChain, LLM Orchestration, SentenceTransformers, Prompt Engineering, Groq (GPT-OSS), Gemini Flash, pgvector
- **Backend & Systems:** FastAPI, Flask, REST APIs, SQLAlchemy, PySide6 / PyQt, Multithreading, Distributed Systems, Linux
- **Databases:** PostgreSQL, MySQL, MongoDB, Pinecone
- **Cloud & DevOps:** Docker, Docker Compose, GitHub Actions, CI/CD, Kubernetes, Tekton, Gunicorn, PyInstaller, Debian Packaging
- **Performance & Edge:** CUDA, GPU-Accelerated Inference, RTSP Streaming, Real-Time Video Processing, Linux SBC Deployment

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science — Shree L.R. Tiwari College of Engineering | Jul 2021 – Jul 2025

CGPA: 8.73 / 10

EXPERIENCE

Software Engineer (R&D) — Blujay Robotics Pvt. Ltd. | Jun 2025 – Present

Mumbai, India

- Built a thermal-camera fire classification pipeline trained on 42K+ fire/non-fire samples, engineering 63 temperature, spatial, intensity-gradient, distance, and temporal-flicker features with a Scikit-learn Gradient Boosting pipeline for real-time fire detection in an AI-enabled industrial fire-response system.
- Developed and deployed a lightweight YOLO-based human detection and pose-estimation pipeline for live RTSP streams, using 17-keypoint pose inference to classify standing, sitting, sleeping, and person-missing states while maintaining <15% GPU utilization for real-time deployment.
- Architected a distributed Python software system across 2 Linux SBC controllers, separating perception, navigation, thermal imaging, device-control, and recovery processes while coordinating real-time inter-controller communication.
- Automated application builds and Linux packaging using GitHub Actions, Python, PyInstaller, and Debian tooling, eliminating a ~2-hour manual release workflow and enabling authorized users to retrieve deployment-ready software without engineer-assisted builds.
- Developed an AES-256-GCM application-layer security module to protect RTSP credentials and sensitive model/application paths in deployed systems, using externalized key management to prevent direct exposure at customer sites.

Software Engineering Intern — Blujay Robotics Pvt. Ltd. | Jan 2025 – Apr 2025

Mumbai, India

- Contributed to the ground-up development of Airavat, GPU-accelerating real-time vision workloads using PyTorch and CUDA across 50+ IP and 21 thermal camera streams, reducing CPU utilization from 80%+ to 35–40% while utilizing approximately 20% GPU.
- Engineered an on-demand RTSP streaming pipeline that initializes camera streams in the backend and renders frames only when requested, automatically releasing cached resources when streams are closed to reduce unnecessary memory and compute utilization.
- Re-engineered Airavat's SCADA visualization and mapping system from Matplotlib to PyQt, supporting site-level visualization of cameras and robotic water monitors while eliminating application crashes and memory spikes from the previous implementation.

PROJECTS

AI-Powered Employee Contribution & Sprint Intelligence Platform

- Built an evidence-grounded AI platform that combines PostgreSQL delivery data with RAG over PRD, TRD, architecture, testing, and release documents to analyze documented employee and sprint contributions without relying on unsupported LLM claims.
- Implemented document ingestion with pypdf/python-docx, deterministic chunking, SentenceTransformers MiniLM embeddings, PostgreSQL + pgvector semantic retrieval, hybrid SQL + vector evidence assembly, bounded context, and source-level citation metadata.
- Integrated LangChain-based LLM orchestration with Groq-hosted GPT-OSS as the primary model and Gemini Flash as fallback, producing structured, citation-grounded project intelligence and contribution narratives; LLM grounding and provider resilience are the system's key USP.
- Developed the application with FastAPI, React/TypeScript, SQLAlchemy, Alembic, Docker Compose, automated backend tests, KPI/workflow services, Kanban workflows, and report-ready structured outputs.

Home Loan Default Risk Prediction

- Developed a stacked ensemble for imbalanced loan-default prediction using LightGBM, XGBoost, and CatBoost, achieving 0.88 ROC-AUC on Home Credit default-risk data.
- Engineered domain-specific financial, external-risk, missingness, and behavioral features, implementing leakage-safe categorical encoding, stratified 5-fold cross-validation, and out-of-fold stacking for robust model training and evaluation.
- Integrated the trained ensemble into an end-to-end application using FastAPI, PostgreSQL, Docker Compose, and GitHub Actions, exposing model inference through REST APIs with automated testing and container builds.

RetailDialogue — Conversational Semantic Search

- Engineered a semantic product-discovery pipeline using SentenceTransformers, Pinecone, FastAPI, and Dialogflow, transforming natural-language queries into embeddings for vector similarity search and retrieval of semantically relevant products.
- Implemented multi-turn conversational commerce workflows for product search, cart management, order placement, and order tracking using Dialogflow intent detection, FastAPI webhook fulfillment, and session-based state management.